SALES ANALYTICS DASHBOARD

Power BI Project Report

Project Type	Business Intelligence and Data Analytics
Primary Tool	Microsoft Power BI Desktop
Project File	Sales_Analytics.pbix
Purpose	Explore sales performance by product, country, category, orders, and quantity.

Prepared as a portfolio document for GitHub. This report describes the dashboard structure and intended analysis; numerical findings must be validated against the source dataset.

1. Executive Summary

The Sales Analytics Dashboard is a Power BI business intelligence project designed to turn sales records into interactive visual summaries. It supports comparisons across products, countries, and categories, while also presenting order-count and quantity-related details. The report can help users explore performance patterns and identify areas for further investigation.

The dashboard is a reporting and exploration tool. Conclusions about the best-performing product, country, or category should only be stated after validating the underlying dataset and the measures used.

2. Project Objectives

- Summarize sales-related metrics in a single dashboard.
- Compare revenue across products and countries.
- Examine revenue contribution by category.
- Review product-level quantity and revenue details.
- Explore order-count patterns and geographical context.
- Build practical skills in Power BI visualization and business reporting.

3. Tools and Technologies

Tool / Skill	Purpose
Microsoft Power BI Desktop	Dashboard layout, interactive visuals, and report exploration.
Sales dataset	Source records for product, country, category, order, quantity, and revenue fields.
Data aggregation	Summarizing measures such as revenue, counts, and quantities.
Business analysis	Interpreting performance comparisons to support decisions.

4. Dashboard Components

Visual	Fields / Measure	Purpose
Card	Count of OrderID	Shows nonblank order ID entries; this is not necessarily the number of unique orders.
Card	Count of Product	Shows nonblank product entries.
Bar chart	Product; Sum of Revenue	Compares revenue across products.
Column chart	Country; Sum of Revenue	Compares revenue across countries.
Donut chart	Category; Sum of Revenue	Shows category-level revenue contribution.
Table	Quantity, Product, Revenue	Provides detailed product-level sales information.
Map	Country, Product	Provides geographical context; a numeric measure would make the map more informative.
Area chart	Product; Count of OrderID	Compares order ID row counts by product; consider a line chart over time if a date field is available.

5. Business Questions This Dashboard Can Answer

- Which products generate the highest and lowest revenue?
- Which countries contribute most to sales revenue?

- How is revenue distributed across product categories?
- Which products account for more order entries or quantities?
- Where might sales teams investigate growth opportunities?

These are analytical questions the dashboard is designed to explore. The current report summary does not establish the actual answers or rankings without a validated source-data analysis.

6. Recommended DAX Measures

The following measures can be added if the referenced columns exist in the model. They use the table name analyzed_sales; adjust the table or column names to match the actual Power BI model.

```
Total Revenue = SUM(analyzed_sales[Revenue])

Total Orders = DISTINCTCOUNT(analyzed_sales[OrderID])

Total Quantity = SUM(analyzed_sales[Quantity])

Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)
```

Important: DISTINCTCOUNT counts unique order IDs. A card using Count of OrderID counts nonblank rows, so the two measures can produce different results when an order has multiple rows.

7. Recommendations for Improvement

1. Add KPI cards for Total Revenue, Unique Orders, Total Quantity, and Average Order Value.
2. Add slicers for Country, Product, Category, and Order Date when those fields are available.
3. Improve the map by using Country as a geographic field and Total Revenue as a measure where supported.
4. Consider a time-series line chart for monthly or daily revenue if a valid date field exists.
5. Format revenue consistently as currency and use clear, descriptive titles.
6. Validate data types, blanks, duplicates, and aggregation settings before publishing.

8. Expected Business Value

Once connected to validated sales data, the dashboard can make comparisons easier, reduce manual reporting effort, and help stakeholders focus on products, categories, and markets that deserve closer review. It should be used as decision support rather than as proof of business performance on its own.

9. Conclusion

The Sales Analytics Dashboard demonstrates foundational Power BI skills: report layout, visual selection, aggregation, and sales-oriented analysis. The next stage is to validate the source data, add well-defined KPIs, improve filtering and time-based analysis, and document verified numerical findings. These improvements will make the project more compelling as a Data Analyst portfolio item.

10. GitHub Repository Checklist

Suggested repository structure:

```
Sales-Analytics-Project/
■■■ Sales_Analytics.pbix
■■■ Sales_Analytics_Project_Report.pdf
■■■ README.md
■■■ data/ (optional; share only if permitted)
```

Before pushing to GitHub:

- Check that the dataset can be shared publicly and contains no sensitive information.
- Add a README describing the objective, visuals, tools, and how to open the PBIX file.
- Do not commit passwords, private connection strings, or confidential business data.
- State that numerical insights are pending validation if you have not calculated them.

Report note: This document describes the dashboard structure and recommended analysis based on the available report definition. Actual sales totals, rankings, and percentages have not been independently verified from the underlying data.